Answers on
Booklet - 13

Taken from OCR Physics A past papers

Abbreviations, annotations and conventions used in the Mark Scheme	/	=	alternative and acceptable answers for the same marking point
	;	=	separates marking points
	NOT	=	answers which are not worthy of credit
	( )	=	words which are not essential to gain credit
	____	=	(underlining) key words which **must** be used to gain credit
	ecf	=	error carried forward

1.

i	$3.0 \times 10^8 = 1.0 \times 10^9 \lambda$	C1	
	$\lambda = 0.30$ m	A1	**allow** 0.3 no SF error
ii	aerial length = $\lambda/2$ = 0.15 m	A1	**ecf (c)(i)**
iii	emitted wave is (plane) polarised	B1	**allow** max signal initially/at 0°
	detecting aerial will receive weaker signal/cos θ component		
	when it is rotated (through angle θ)/AW	B1	
	signal falls to zero at 90°	B1	
	and then rises to max again at 180°		**max 3 marks** from 4 marking points

2.

(a)	**(i)**	light-dependent resistor \ LDR	B1
(a)	**(ii)**	Resistance of **X** decreases (as light intensity is increased)	B1
		The current (in the circuit) increases	B1
(a)	**(iii)**	The current is halved.	B1

3.

(a)(i)	Coulomb / C	(Allow Ampere second / As)	B1
(ii)	Voltmeter		B1
(b)(i)	P = VI	(Allow other variant)	C1
	V = 36 / 3.0		C1
	p.d. = 12 (V)		A1
(ii)	E = 36 x 600		C1
	energy = $2.1(6) \times 10^4$ (J) ≈ 2.2×10^4 (J)		A1
(iii)	$\Delta Q = I\Delta t$ / Q = It	(Allow other variant)	C1
	$\Delta Q = 3.0 \times 600$		C1
	charge = 1.8×10^3 (C)		A1
(iv)	N = $1.8 \times 10^3 / 1.6 \times 10^{-19}$	(Possible ECF)	C1
	number = $1.1(3) \times 10^{22} \approx 1.1 \times 10^{22}$		A1

4.

(a) diagram showing		
laser/light source placed directly behind double slit AND screen placed in front of slits	B1	[1]
{single slit NOT required; no labelling required}		
(i) *D*: allow any value between 30cm and 10m	B1	
(ii) *a*: allow any value between 0.1mm and 2mm	B1	[2]
(b) (i) evidence of good practice: i.e distance for nx measured e.g. 5x = 18mm	C1	
x = 3.6 mm (OR 3.5 OR 3.7)	A1	[2]
{x = 3.4, 3.8, 3.9, 4.0, or 4 mm, implying x is directly measured, and score 1 mark)		
(ii) for O path difference = 0	B1	
for A path difference = 3(λ)	B1	
for B path difference = 1.5(λ)	B1	[3]
(c) recall of λ= ax/D OR x =λD/a OR x $\propto \lambda$	B1	
λ is smaller for blue light (than red light) hence x is SMALLER (WTTE)	B1	[2]

5.

(i)	INTERFERENCE (OR SUPERPOSITION)	B1	[1]
(ii)	Any 2 points from:		
	equal distance travelled by waves from both slits (OR zero path diff.)}		
	waves meet in phase	B1+B1	
	constructive interference occurs		
(iii)	idea that path difference/phase difference changes implied in 1. or 2}.	B1	
OR	correct reference to destructive AND constructive interference		
	1. low reading at A because path difference is an odd number of ½λs	B1	
OR	180° out of phase (or in antiphase)		
	2. high reading at B when path difference is a whole number of λs (not 0)	B1	[3]
OR	waves meet in phase (allow zero or 2π phase difference		
(iv)	**FEWER** (highs and lows) (OR more spread out)	M1	
	any valid explanation: e.g reference to $x \propto 1/a$ OR $x=\lambda D/a$ (not $\lambda = ax/D$)	A1	[2]
OR	since wave sources closer together, further distance needed along the arc to create the longer path difference of λ. (WTTE)		

6.

(a)	$R = R_1R_2 / R_1 + R_2$ \ $1/R = 1/R_1 + 1/R_2$		C1
	$R = 6.0 \times 2.0 / (6.0 + 2.0)$ \ $1/R = 1/2.0 + 1/6.0$		C1
	$R = 1.5\ (\Omega)$		A1
(b)	$P = VI$ \ I^2R \ V^2/R		C1
	$P = 12^2 / 2.0$		C1
	$P = 72$	(-1 if P = 24 (W) for coil Y)	A1
	unit: W \ Js^{-1}		B1

7.

(a)	Line crosses 'y-axis' at 1.4 (V) / $V = E$ or 1.4(V) when $I = 0$ $V = E - Ir$, since $I = 0$ (Hence $V = E$ or 1.4(V))		**B1**
(b)(i)	(Graph extrapolated to give) current = 2.0 (A) (Allow tolerance ± 0.1A)		**B1**
(b)(ii)	$E = I_{(max)}\, r$ $(r = \frac{1.4}{2.0})$ $r = 0.7(0)\ (\Omega)$	gradient = r (Ignore sign) (Attempt made to find gradient) $r = 0.7(0)\ (\Omega)$ (Possible ecf)	**C1** **A1**
(b)(iii)	(excessive) heating of <u>cell</u> / energy wasted <u>internally</u> / cell might 'explode' / <u>cell</u> goes 'flat' (quickly)		**B1**

8.

(a) **(i)** ANY 3 correct phenomena from REFLECTION, REFRACTION, INTERFERENCE, SUPERPOSITION, DIFFRACTION, (allow transfer energy) B2 **[2]**

3 correct scores 2 marks, 2 correct scores 1 mark otherwise zero

(ii) POLARISATION B1 **[1]**

(b) **(i)** it consists of nodes and antinodes / it does not transfer energy (WTTE) B1
formed by two identical waves travelling in opposite directions (WTTE) B1
(microwaves leaving transmitter) interfere (with reflected waves) (WTTE)B1 **[3]**
{allow superimpose/interact/cancel out/reinforce for interfere}

(ii) 1.wavelength of the microwaves = 2 x 1.4 = 2.8 cm B1 **[1]**

2. speed of microwaves in air = 3 x108 m/s OR c M1
frequency = 3x108 /2.8 x10-2 (allow ecf) = 1.07 x 1010 Hz A1 **[2]**

(iii) Place a metal grid {allow "Polaroid"} (between T and D) and rotate
(or place at 900) OR rotate grid/transmitter/detector B1
this causes minm/zero signal (WTTE) B1 **[2]**

9.

(a) transverse waves: vibrations 90° to wave direction (WTTE) **AND**
longitudinal waves: vibrations parallel to wave direction (WTTE) B1
example of transverse waves: e.g light B1
example of longitudinal waves: e.g sound B1 [3]

(b) ***similarity***: anything valid: e.g. both have wavelength/frequency/vibrations B1
difference: anything valid: e.g. B1 [2]
no transfer of energy in standing waves
standing waves have nodes (&/or antinodes)
neighbouring pts vibrate in phase in standing waves but have a phase diff. in progressive

(c) (i) node = point of no movement/zero displacement (allow amplitude) B1
(ii) antinode = point of maximum movement/max displacement (amplitude) B1 [2]
{for diagrams with no words maximum of 1 mark}

(d) wavelength = 4x0.12 = **0.48**m B1 [1]

(e) (labelled) diagram of <u>valid</u> arrangement: i.e for **longitudinal** waves B1
wave source stated: e.g. tuning fork/loud-speaker/oscillator/hand moving slinky- B1

explanation of how the standing wave is formed: e.g.
waves leaving wave sources **interfere/superpose** with reflected waves B1

*{allow this mark for correct **procedure** used to produce **longitudinal** the standing wave}*

node (N) AND antinodes (A) shown in correct positions and labelled anywhere B1 [4]

{all ecf from cand's defns in (c)}

{answers in terms of transfer waves may score a max of 3 marks}

10.

Any four from: B1 × 4

1 (As temperature increases) the resistance of the thermistor / **T** decreases
2 The total resistance decreases (Possible ecf)
3 The current increases (in the circuit) (Possible ecf)
4 The (voltmeter) reading increases / voltage across **R** increases (Possible ecf)
5 The voltage across the thermistor / **T** decreases (Possible ecf)
6 Correct use of the potential divider equation / comment on the 'sharing' of voltage / correct use of $V = IR$

11.

(a) No current (in circuit) / 'open' circuit / p.d. between **X** and **Y** is 5.0 V B1

(b) $V = \frac{R_2}{R_1 + R_2} \times V_0$ / $\frac{V_1}{V_2} = \frac{R_1}{R_2}$ / $I = \frac{3.4}{168}$ (= 2.02×10^{-2} mA) C1

$3.4 = \frac{168}{168 + R} \times 5.0$ / $\frac{1.6}{3.4} = \frac{R}{168}$ / $R = \frac{1.6}{2.02 \times 10^{-2}}$ C1

resistance ≈ 79 (kΩ) (Total resistance of 250 kΩ scores 2/3) A1

12.

(a) Sum of currents = zero (at junction) / sum of currents in = sum of currents out (at junction) — B1
Charge is conserved. — B1

(b)(i)

$V = V_0 \times R_2 / (R_1 + R_2)$	$R = V/I$ and $R = R_1 + R_2$	C1
$V = 6.0 \times 400 / (1200 + 400)$	$I = 6.0 / 1600 = 3.75 \times 10^{-3}$ (A)	
	$V = 3.75 \times 10^{-3} \times 400$	C1
$V = 1.5$(V)	$V = 1.5$(V)	A0
(Answer of 4.5 (V) scores 1/ 2)		

(ii)

1. $R_D = 1.5 / 0.1 = 15\ (\Omega)$ — C1
$R = R_1R_2 / (R_1 + R_2)$ / $1/R_T = \sum 1/R_1$ — C1
$R = 400 \times 15 / (400 + 15) = 14.5 = 15\ (\Omega)$ (possible e.c.f) — A1

2.

$V \approx 6.0 \times 15 / (1200 + 15)$	$I \approx 6.0 / 1200 + 15 \approx 4.98 \times 10^{-3}$ (A)	C1
$V \approx 0.07$ (V)	$V \approx 4.98 \times 10^{-3} \times 15 = 0.07$ (V)	A1
(Answer of 5.93 (V) scores 1/ 2)		

3. Resistance of device is small(er) / current in device is small(er) / p.d. across device is small(er) — B1

13.

(a) Any two from: — B1 × 2
Travel at the speed of light / 3×10^8 (ms^{-1}) (in vacuum)
Can travel in a vacuum / space
Transverse waves / Can be polarised
Consist of oscillating electric and magnetic fields
May be diffracted / reflected / refracted etc.

(b)(i) Quantum of energy / radiation
OR 'packet' / 'bundle' / 'lump' of energy — B1
(Do not allow 'E =hf' here)

(ii) $E = 1.0 \times 10^6 \times 1.6 \times 10^{-19}$
use of 10^6 factor in answer — C1
use of 1eV = 1.6×10^{-19} (J) in answer — C1
$E = 1.6 \times 10^{-13}$ (J) — A0

(iii) E = hf (Allow this if given in **(b)(i)**) (Allow other variants) — C1
$f = 1.6 \times 10^{-13} / 6.63 \times 10^{-34}$ — C1
$f = 2.4(1) \times 10^{20}$ — A1
unit: hertz / Hz (Allow hz / HZ - as BOD) — B1

(iv) $\lambda = 3.0 \times 10^8 / 2.4(1) \times 10^{20}$ — C1
$\lambda = 1.2(4) \times 10^{-12}$ (m) (Possible e.c.f) — A1

(v) Principal radiation: γ-rays / radiation. — B1

14.

(a)(i)	Photoelectric (effect)	B1
(a)(ii)	10^{-9} (m) ≤ wavelength ≤ 4×10^{-7} (m)	B1
(b)(i)	(Minimum) energy needed to free an electron /an electron to escape (from the metal surface)	B1
(b)(ii)	speed of light / 3×10^{8} (m s^{-1}) / c	B1
(b)(iii)1.	$hf = \phi + KE_{(max)}$ (Allow any subject)	C1
	$KE_{max} = 2.8 - 1.1 = 1.7$ (eV)	C1
	$KE_{max} = 1.7 \times 1.6 \times 10^{-19}$	
	$KE_{max} = 2.7 \times 10^{-19}$ (J)	A1
(b)(iii)2.	$\frac{1}{2} mv^2 = 2.7 \times 10^{-19}$ (Possible ecf)	C1
	$v = \sqrt{\frac{2 \times 2.7 \times 10^{-19}}{9.1 \times 10^{-31}}}$	
	$v = 7.7 \times 10^{5}$ (m s^{-1})	A1
(b)(iv)	No change (because the energy of the photon remains the same)	B1

15.

(i)	The <u>minimum</u> frequency needed to free an electron (from the surface of a metal)	B1
(ii)1	Line extended intersects (the f axis at) this value / At this frequency, $E_k = 0$	B1
(ii)2	(ϕ =) $h \times 5.0 \times 10^{14}$ / (ϕ =) $6.63 \times 10^{-34} \times 5.0 \times 10^{14}$	C1
	work function energy = 3.3×10^{-19} J	A1
(iii)1	The gradient / slope of the line is the same	B1
	The gradient is equal to h / independent of the metal	B1
(iii)2	The line is shifted to the right	B1
	The threshold frequency is greater (AW)	B1